Assignment 7(CSB24084)

```
from typing import List, Dict, Set
from collections import defaultdict
from functools import reduce
import math


logs: List[Dict[str, object]] = [
    {"user": "CSB24001", "action": "YouTube", "duration": 30.5},
    {"user": "CSB24002", "action": "Instagram", "duration": 45.0},
    {"user": "CSB24001", "action": "VSCode", "duration": 120.0},
    {"user": "CSB24003", "action": "YouTube", "duration": 25.0},
    {"user": "CSB24002", "action": "Chrome", "duration": 60.0},
    {"user": "CSB24001", "action": "Instagram", "duration": 15.5},
]


def total_time_per_user(logs: List[Dict[str, object]]) -> Dict[str, float]:
    totals: defaultdict[str, float] = defaultdict(float)
    for log in logs:
        totals[log["user"]] += float(log["duration"])
    return dict(totals)


def most_active_users(logs: List[Dict[str, object]], k: int) -> List[str]:
    totals = total_time_per_user(logs)
    return [
        user
        for user, _ in sorted(
            totals.items(),
            key=lambda item: item[1],
            reverse=True
        )[:k]
    ]


def unique_actions(logs: List[Dict[str, object]]) -> Set[str]:
    return {log["action"] for log in logs}


def total_activity_time(logs: List[Dict[str, object]]) -> float:
    return reduce(lambda acc, log: acc + float(log["duration"]), logs, 0.0)


if __name__ == "__main__":
```

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    n = len(logs)

    print("Total Time Per User:")
    totals = total_time_per_user(logs)
    print(totals)

    print("\nMost Active 2 Users:")
    top_users = most_active_users(logs, 2)
    print(top_users)

    print("\nUnique Actions:")
    print(unique_actions(logs))

    print("\nTotal Activity Time (All Users):")
    print(total_activity_time(logs))

    u = len(totals)

    print("\nComplexity Analysis:")

    time_complexity_value = u * math.log2(u) if u > 0 else 0
    print(f"Time Complexity for Top K Users:")
    print(f"O(n + u log u)")
    print(f"Here n = {n} (total logs), u = {u} (unique users)")
    print(f"Approximate operations = {int(n + time_complexity_value)}")

    print("\nSpace Complexity:")
    print("O(u) for storing intermediate user totals dictionary")
    print(f"Intermediate dictionary size = {u}")
```

Output

```
Total Time Per User:
{'CSB24001': 166.0, 'CSB24002': 105.0, 'CSB24003': 25.0}

Most Active 2 Users:
['CSB24001', 'CSB24002']

Unique Actions:
{'Instagram', 'Chrome', 'VSCode', 'YouTube'}

Total Activity Time (All Users):
296.0
```

Complexity Analysis:
Time Complexity for Top K Users:
$O(n + u \log u)$
Here n = 6 (total logs), u = 3 (unique users)
Approximate operations = 10

Space Complexity:
$O(u)$ for storing intermediate user totals dictionary
Intermediate dictionary size = 3